

**MS:**

$$2.1 \quad x^1 = \boxed{x}$$

$$2.2 \quad -a \cdot (-b)^2 = \boxed{-ab^2}$$

$$2.3 \quad \sum_{i=1}^4 x_i = \boxed{x_1 + x_2 + x_3 + x_4}$$

$$2.4 \quad \prod_{m=6}^9 x_m = \boxed{x_6 \cdot x_7 \cdot x_8 \cdot x_9}$$

$$2.5 \quad 4! = \boxed{4 \cdot 3 \cdot 2 \cdot 1}$$

$$2.6 \quad z^4 = \boxed{z \cdot z \cdot z \cdot z}$$

$$2.7 \quad \sqrt[2]{9} = \boxed{\pm 3}$$

$$2.8 \quad \sqrt[3]{27} = \boxed{3}$$

$$2.9 \quad \left[ \frac{3(2-4)}{2+3} \right]^3 = \boxed{\left( -\frac{6}{5} \right)^3 = -\frac{216}{125}}$$

$$2.15 \quad (\text{a}) \quad xz + yz = \boxed{z(x + y)}$$

$$(\text{b}) \quad mn + ln - pn = \boxed{n(m + l - p)}$$

$$(\text{c}) \quad zxy - 2xy = \boxed{xy(z - 2)}$$

$$(\text{d}) \quad (z + x) \times y \times \frac{1}{z} = \boxed{\frac{(z+x) \cdot y}{z}}$$

$$2.16 \quad \frac{2x^2+20x+50}{2x^2-50} = \frac{2(x^2+10x+25)}{2(x^2-25)} = \frac{(x+5)(x+5)}{(x+5)(x-5)} = \boxed{\frac{x+5}{x-5}}$$

$$2.17 \quad \frac{5+17x+4x+7}{42x} = \frac{21x+12}{42x} = \boxed{\frac{1}{2} + \frac{2}{7x}}$$

$$2.18 \quad \frac{2g+13}{3g} + \frac{4g-5}{4g} = \frac{8g+52+12g-15}{12g} = \boxed{\frac{20g+37}{12g}}$$

$$2.19 \quad -7\theta^2 + 21\theta - 14 = -7(\theta^2 - 3\theta + 2) = \boxed{-7(\theta - 2)(\theta - 1)}$$

$$2.20 \quad (2x - 3)(5x + 7) = 10x^2 + 14x - 15x - 21 = \boxed{10x^2 - x - 21}$$

$$2.21 \quad q^2 - 10q + 9 = \boxed{(q - 9)(q - 1)}$$

$$2.22 \quad \frac{\beta - \alpha}{\alpha^2 - \beta^2} = \frac{-(\alpha - \beta)}{\alpha^2 - \beta^2} = -\frac{(\alpha - \beta)}{(\alpha + \beta)(\alpha - \beta)} = \boxed{\frac{-1}{\alpha + \beta}}$$

$$2.23 \quad 15\delta + 45 - 6\delta = 36$$

$$9\delta = -9 \implies \boxed{\delta = -1}$$

$$2.24 \quad 0.30\Omega + 0.05 = 0.25$$

$$0.30\Omega = 0.2 \implies \boxed{\Omega = \frac{2}{3}}$$

$$2.25 \quad 11 = (y+1)2 + (6y-12y)\frac{7}{2y}$$

$$\implies 2y + 2 + 21 - 42 = 11 \implies 2y - 19 = 11 \implies \boxed{y = 15}$$

$$2.26 \quad -4x^2 + 64 = 8x - 32$$

$$\implies 4x^2 + 8x - 96 = 0 \implies (x+6)(x-4) = 0 \implies \boxed{x = -6}, \boxed{x = 4}$$

$$2.27 \quad x^2 + 14x - 14 = 0$$

$$\implies \underbrace{x^2 + 14x = 14}_{\substack{\text{Move constant term} \\ -14 \text{ to right side}}} \implies \underbrace{x^2 + 14x + 49 = 63}_{\substack{\text{Add and subtract } 7^2=49 \\ \text{to complete the square}}} \implies \underbrace{(x+7)^2 = 63}_{\text{Perfect square}} \implies \underbrace{x = -7 \pm \sqrt{63}}_{\text{Take square root \& solve}}$$

$$2.28 \quad \frac{1}{3}y^2 + \frac{2}{3}y - 16 = 0$$

$$\implies y^2 + 2y = 48 \implies y^2 + 2y + 1 - 1 = 49 \implies (y+1)^2 = 49 \implies y+1 = \pm 7 \implies \boxed{y = 6}, \boxed{y = -8}$$

$$2.29 \quad 2x^2 + 5x - 7 = 0$$

$$\implies x = \frac{-5 \pm \sqrt{25 + 56}}{4} = \frac{-5 \pm \sqrt{81}}{4} = \boxed{x = 1}, \boxed{x = -\frac{7}{2}}$$

$$2.30 \quad ax^2 + bx + c = 0$$

$$\implies x^2 + \frac{b}{a}x = -\frac{c}{a} \implies \left(x + \frac{b}{2a}\right)^2 = \left(\frac{b}{2a}\right)^2 - \frac{c}{a} \implies \left(x + \frac{b}{2a}\right)^2 = -\frac{c}{a} + \left(\frac{b^2}{4a^2}\right)$$

$$\implies \left(x + \frac{b}{2a}\right)^2 = \frac{-4ac + b^2}{4a^2} \implies x + \frac{b}{2a} = \pm \frac{\sqrt{b^2 - 4ac}}{2a} \implies x = \boxed{\frac{-b \pm \sqrt{b^2 - 4ac}}{2a}}$$

$$2.31 \quad -\delta > \frac{\delta+4}{7} \implies -7\delta > \delta + 4 \implies -8\delta > 4$$

$$\implies \delta < \boxed{-\frac{1}{2}}$$

$$3.2 \quad f(x) = x^2 + 2, \quad g(x) = \sqrt{x-4}$$

$$g(f(x)) = \sqrt{(x^2 + 2) - 4} = \boxed{\sqrt{x^2 - 2}}$$

$$3.3 \quad f(g(x)) = (\sqrt{x-4})^2 + 2 = x - 4 + 2 = \boxed{x - 2}$$

$$\implies f(g(x)) \neq g(f(x))$$

$$3.4 \quad y = f(x) = 5x - 2 \Rightarrow x = \frac{y+2}{5}$$

$$\Rightarrow f^{-1}(x) = \boxed{\frac{x+2}{5}}$$

$$3.5 \quad x^{-2} \cdot x^3 = \boxed{x}$$

$$3.6 \quad (b \cdot b \cdot b) \cdot c^{-3} = \boxed{\left(\frac{b}{c}\right)^3}$$

$$3.7 \quad ((qr)^\gamma)^\delta = \boxed{(qr)^{\gamma\delta}}$$

$$3.8 \quad x^{1/2} \cdot x^{1/5} = \boxed{x^{\frac{7}{10}}}$$

$$3.9 \quad \ln(3x) - 2\ln(x+2) = \boxed{\ln\left(\frac{3x}{(x+2)^2}\right)}$$

### From Sydsaeter and Hammond

1.7.3

(a)  $|3 - 2x| = 5$

$$3 - 2x = 5 \quad \text{or} \quad 3 - 2x = -5$$

$$\Rightarrow \boxed{x = -1} \quad \text{or} \quad \boxed{x = 4}$$

(b)  $|x| \leq 2$

$$\boxed{-2 \leq x \leq 2}$$

(c)  $|x - 2| \leq 1$

$$-1 \leq x - 2 \leq 1$$

$$\boxed{1 \leq x \leq 3}$$

(d)  $|3 - 8x| \leq 5$

$$-5 \leq 3 - 8x \leq 5$$

$$-8 \leq -8x \leq 2$$

$$\boxed{1 \geq x \geq -\frac{1}{4}}$$

(e)  $|x| > \sqrt{2}$

$$\boxed{x < -\sqrt{2}} \quad \text{or} \quad \boxed{x > \sqrt{2}}$$

(f)  $|x^2 - 2| \leq 1$

$$-1 \leq x^2 - 2 \leq 1$$

$$1 \leq x^2 \leq 3 \Rightarrow \sqrt{1} \leq |x| \leq \sqrt{3}$$

$$\Rightarrow \boxed{-\sqrt{3} \leq x \leq -1} \quad \text{or} \quad \boxed{1 \leq x \leq \sqrt{3}}$$

Cont.

## 2.4.1

(a)

$$x - y = 5$$

$$x + y = 11$$

$$\implies 2x = 16 \implies x = \boxed{8}$$

$$x + y = 11 \implies y = \boxed{3}$$

(b)

$$4x - 3y = 1 \implies 12x - 9y = 3$$

$$2x + 9y = 4 \implies 2x + 9y = 4$$

$$\implies 14x = 7 \implies x = \boxed{\frac{1}{2}}$$

$$\implies \left(2\frac{1}{2}\right) + 9y = 4 \implies \boxed{y = \frac{1}{3}}$$

(c)

$$3x + 4y = 2.1$$

$$5x - 6y = 7.3$$

$$\implies x = \frac{2.1 - 4y}{3}$$

$$5\left(\frac{2.1 - 4y}{3}\right) - 6y = 7.3$$

$$\implies \frac{10.5 - 20y - 18y}{3} = 7.3$$

$$\implies -38y = 11.4 \implies \boxed{y = -0.3}$$

$$3x + 4(-0.3) = 2.1 \implies \boxed{x = 1.1}$$

4.2.1. Let  $f(x) = x^2 + 1$ 

(a)

$$f(0) = \boxed{1}, \quad f\left(\frac{1}{2}\right) = \boxed{\frac{5}{4}}, \quad f(\sqrt{2}) = \boxed{3}$$

(b) (i)  $f(x) = f(-x)$ 

$$f(x) = x^2 + 1$$

$$f(-x) = x^2 + 1 = f(x) \quad \boxed{\forall \ x}$$

$$(ii) f(x+1) = f(x) + f(1)$$

$$f(x+1) = (x+1)^2 + 1 = x^2 + 2x + 1 + 1 = x^2 + 2x + 2$$

$$f(x) + f(1) = x^2 + 1 + 1 + 1 = x^2 + 3$$

$$\text{Set } f(x+1) = f(x) + f(1)$$

$$x^2 + 2x + 2 = x^2 + 3$$

$$2x = 1 \implies x = \boxed{\frac{1}{2}}$$

$$(iii) f(2x) = 2f(x)$$

$$f(2x) = 4x^2 + 1$$

$$2f(x) = 2(x^2 + 1) = 2x^2 + 2$$

$$4x^2 + 1 = 2x^2 + 2$$

$$2x^2 = 1 \implies x = \boxed{\pm\sqrt{\frac{1}{2}}}$$

4.3.3

|        |   |   |   |    |    |
|--------|---|---|---|----|----|
| $x$    | 0 | 1 | 2 | 3  | 4  |
| $g(x)$ | 5 | 3 | 1 | -1 | -3 |

4.3.4

|        |    |    |    |    |    |   |   |
|--------|----|----|----|----|----|---|---|
| $x$    | -2 | -1 | 0  | 1  | 2  | 3 | 4 |
| $h(x)$ | 5  | 0  | -3 | -4 | -3 | 0 | 5 |

4.10.2 Solve the following equations for  $x$ :

$$(a) 3^x = 8$$

$$x \ln 3 = \ln 8 \implies \boxed{x = \frac{\ln 8}{\ln 3}}$$

$$(b) \ln x = 3$$

$$\boxed{x = e^3}$$

$$(c) \ln(x^2 - 4x + 5) = 0$$

$$x^2 - 4x + 4 = 0 \implies (x - 2)^2 = 0 \implies \boxed{x = 2}$$

$$(d) \ln[x(x - 2)] = 0$$

$$x(x - 2) = 1 \implies x^2 - 2x - 1 = 0$$

$$\boxed{x = 1 \pm \sqrt{2}}$$

Cont.

$$(e) \frac{x(x+3)}{x^2+1} = 0$$

$$x = 0, x = -3$$

$$(f) \ln(\sqrt{x} - 5) = 0$$

$$\sqrt{x} - 5 = 1 \implies \sqrt{x} = 6 \implies x = 36$$

### Additional Questions:

$$(a) \quad (i) \quad c(p(t)) = 0.5(10 + 0.1t^2) + 1 = 5 + .05t^2 + 1 = 6 + .05t^2$$

$$(ii) \quad c(p(t)) = 6.8 = 6 + .05t^2 \implies .8 = .05t^2 \implies t = \left(\frac{.8}{.05}\right)^{\frac{1}{2}} = 4$$

$$(b) \quad \epsilon_D^P = -\frac{1}{4} \text{ and } \epsilon_S^P = \frac{1}{2}$$

Generic supply and demand:  $Q_D = a - bP$ ;  $Q_S = c + dP$ , where a, b, c, and d are constants.

$$\text{Recall } \epsilon_D^P = \frac{\Delta Q}{\Delta P} \cdot \frac{P}{Q}$$

$$\implies \epsilon_D^P = -b \cdot \frac{P}{Q} \implies -\frac{1}{4} = -b \frac{20}{100} \implies b = \frac{5}{4}$$

$$\implies Q_D = a - \frac{5}{4} \cdot P \implies 100 = a - \frac{5}{4} \cdot 20 \implies a = 125$$

$$\implies Q_D = 125 - \frac{5}{4}P$$

$$\text{For supply, } \epsilon_S^P = d \cdot \frac{P}{Q} \implies \frac{1}{2} = d \cdot \frac{20}{100} \implies d = \frac{5}{2}$$

$$100 = c + \frac{5}{2}(20) \implies c = 50$$

$$\implies Q_S = 50 + \frac{5}{2}P$$